CHM345: Modeling Earth's Climate

JupyterLite Course Site — Instructor Guide

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Site: `https://william-pfalzgraff.github.io/CHM345_FA26/` Repository: `https://github.com/william-pfalzgraff/CHM345_FA26`

Verified against the pinned versions in `requirements.txt`: jupyterlite-core 0.8.3, jupyterlite-pyodide-kernel 0.8.5, ipympl 0.10.0. The Pyodide runtime ships numpy 2.4.6, matplotlib 3.10.8, pandas 3.0.2, h5py 3.13.0.

1 How the site works (30-second model)

- **`content/` is** the students' file browser. On every push to `main`, GitHub Actions rebuilds the site and deploys it. What students open in their browser is a *seed copy*; their edits live in their own browser's storage, layered on top. Anything they haven't touched always mirrors `content/`.
- **A student's local copy shadows the seed.** Once someone opens a notebook, pushing edits to that same file won't reach them. Publish a `_v2` filename to fix a notebook people have started.
- The pinned `requirements.txt` freezes every version for the semester. Do not upgrade anything until winter break.

2 Weekly publishing rhythm

1. Author/edit the week's notebook wherever you like (drafts stay *out* of this repository — everything in `content/` is world-readable the moment you push).
2. When final, copy the week folder in: `content/Week_05a.WhateverItIs/`.
3. Preview locally (Section 3), then:

```
git add content/Week_05a.WhateverItIs
git commit -m "Publish week 5a"
git push
```

4. About two minutes later it's live. Verify **in a private/incognito window** — incognito always shows exactly what's published, with no local shadow.

To *unpublish*: `git rm -r content/<folder>`, commit, push. It vanishes from the seed (students who already opened it keep their local copy).

3 Previewing and iterating

Local preview (identical to production):

```
mkdir -p content   # git drops the dir entirely if it's empty
~/miniforge3/envs/chm345lite/bin/jupyter lite build --contents content --output-dir _output
cd _output && ~/miniforge3/envs/chm345lite/bin/python -m http.server 8898
```

then open `http://127.0.0.1:8898` — same Pyodide, same wheels, same everything.

On the live site: iterate freely *before* sharing the link. Test in incognito each time, or delete your local copy of the file in the JupyterLite file browser (that restores the fresh seed). The "can't update an opened notebook" rule is per-browser-profile, not global.

4 Adapting each notebook for JupyterLite (checklist)

1. **Interactive plots:** the install cell must run **before any code cell that imports matplotlib** (once matplotlib is imported, the widget backend can't register until kernel restart). Convention: the notebook's opening markdown/title cell stays first — markdown doesn't affect this — then these two code cells, then the usual imports:

   ```
   %pip install -q ipympl
   ```

   ```
   %matplotlib widget
   import numpy as np
   import matplotlib.pyplot as plt
   ...
   ```

   Wheels are bundled in the site (`pypi/`), so `%pip install` works offline and always gets the pinned version. Delete any `%matplotlib inline` lines.

2. **MECLib weeks:** copy `MECLib.py` into that week's folder (visible to students, same folder as the notebook) and change the collaborator's `import meclib.cl as cl` to `import MECLib as cl`.

3. **Scenario files:** notebooks reference `../../ScenarioLibrary/...`, which escapes the student drive. Keep a `ScenarioLibrary/` folder at the `content/` root and change paths to `../ScenarioLibrary/...`. Ship pre-built `.pkl` fixtures for later weeks so a student who lost their Week-4 output isn't stuck. (Verified: the existing course `.pkl` files load fine in Pyodide.)

4. **Week 11 (ClimateStats):** the NOAA `https://gml.noaa.gov/...` downloads will *not* work in the browser (CORS). Use the local `brw/` data folder that already exists next to the notebook and point `pd.read_csv` at `brw/met_brw_insitu_1_obop_hour_YYYY.txt`. The folder has 1977–2000 and 2017–2021; the notebook currently asks for 2020–2025 — reconcile the years.

5. **Images:** `http://` images (webspace.pugetsound.edu) are blocked on an HTTPS site. Save them into the week folder and use relative paths.

6. **HDF5:** works for arrays/scalars (h5py, h5io). One limitation: reading *DataFrames stored inside* HDF5 (the 2024 scenario format) needs pytables, which Pyodide doesn't have. Use pickle for anything with a DataFrame — which is what the 2026 notebooks already do.

5 Hard rules

- **Never** put instructor/solution notebooks anywhere in this repository. The `.gitignore` blocks `*instructor version*` paths as a backstop, but the real rule is: solutions live outside `~/Desktop/Courses/CHM345/JupyterLite` entirely.
- Don't push half-finished notebooks to `main` — pushing = publishing.
- Don't change `requirements.txt` or `pypi/*.whl` mid-semester.

6 Student-facing habits — the week-1 talk

Where your work lives. Student edits are stored by the browser (IndexedDB), keyed to the site address. Their "Jupyter" is: *same computer + same browser + same profile.* Within that combination, work persists indefinitely — across closed tabs, restarts, and weeks away.

What wipes or "loses" work — warn students explicitly:

- **Clearing browsing data.** Strictly it's "cookies and site data" (not the image cache) that deletes work, but students won't distinguish — treat any clear-my-browser action as fatal to unsaved-elsewhere work.
- **Switching browser, profile, or computer** — work doesn't follow. It's not deleted (it's still in the original browser), but they'll see a fresh copy and panic. Lab machines that reset profiles on logout lose work every time — students on lab computers *must* download at the end of each session.
- **Safari auto-evicts** site storage after about 7 days without a visit. Weekly class use mostly protects them; a break doesn't. Recommend Chrome or Firefox.
- **Nearly-full disks** can trigger the browser to evict site storage.
- **Incognito/private windows:** everything evaporates when the window closes. Say it explicitly: *never do coursework in a private window.*
- Deleting a file in the Jupyter file browser resets that file to the published version (this is also the fix if they've mangled a notebook and want a fresh start).

The safety net: download your notebook at the end of every session. This is self-enforcing here, since submitting to Gradescope requires downloading — every submission doubles as an off-browser backup. Recovery is the Upload button. A system-check cell that verifies file storage is active (`os.getcwd()` starts with `/drive`) is in `tests/smoke/Welcome.ipynb` — paste it into your own week-1 notebook; run it if anyone's saves vanish.

If plots or `%matplotlib widget` misbehave: Kernel → Restart, then **Run All** (not "continue where you were"). Two things students should understand: (1) if matplotlib gets imported before the `%pip install -q ipympl` cell has run, `%matplotlib widget` errors with "'widget' is not a recognised GUI loop or backend name" — the backend list is frozen at first matplotlib import; (2) restarting wipes `%pip`-installed packages along with everything else, so the install cell must be re-run after every restart. It's instant (wheels are served from the course site) — but "restart, then run from the top" is the reflex to drill, and it fixes this error every time.

7 Semester watch list (from the August 2026 notebook audit)

- **Week 4 prep — MECLib: resolved August 2026.** Steven's current `MECLib.py` (in `~/Desktop/Past MEC materials/`) has every function the 2026 notebooks call, is pickle-based (no h5io), and passed a full in-browser model run (Cambio2, 599 steps). Before publishing it: strip the leaked `### END SOLUTION` marker (near line 233). Known latent bug, not course-blocking: `Diagnose_Delta_T_from_albedo` reads underscore-style keys (`'albedo_sensitivity'`) that `CreateClimateParams` never creates (it uses spaces) — KeyErrors if ever called; worth reporting to Steven. A few functions lack docstrings (`run_Cambio`, `CS_list_plots`, `CS_list_compare`).

- **Week 4:** create `content/ScenarioLibrary/` and fix `../../` → `../` paths (9 notebooks). Generate fixture pickles — `Peaks_in_2040_LTE.pkl` (weeks 6, 11, 13) doesn't exist yet; only `Peaks_in_2040.pkl` and `RCP4_5.pkl` do.

- **Week 6:** delete `%matplotlib inline` from Cambio1.0 (kills the widget backend). The hardcoded `index_of_2003` value is tied to the fixture's time grid — regenerating pickles with a different `nsteps` silently breaks it (wrong answers, not errors).

- **Weeks 3/4c/7/10:** vendor the four `http://webspace.pugetsound.edu` images (mixed content = blocked) plus the two hotlinked HTTPS images in Week 10b.

- **Weeks 10/13:** add `pint` to the install cell; warn students its first import takes a few seconds (not hung).

- **Week 11:** NOAA downloads fail in-browser (CORS). Use the local `brw/` folder (17 MB, already next to the notebook) — but reconcile years: notebook wants 2020–2025, folder has 1977–2000 and 2017–2021.

- **Every week:** copy only the needed files into `content/` — source folders contain hidden `.hdf5`/`.pkl` dotfiles, a 2.9 MB `.ClimateStats.ipynb` old draft, `__pycache__`, checkpoints, `.DS_Store`. And remember several student notebooks (04a, 11a, 11b, 13b) NameError on Restart-&-Run-All *by design* until blanks are filled — word the closing instruction accordingly.

- **Never regenerate student-facing pickle files casually** — pickles are tied to the (frozen) pandas version. Safe only because nothing gets upgraded.

8 Local machinery reference

- Build environment: conda env `chm345lite` (`~/miniforge3/envs/chm345lite`).
- `tests/smoke/` (gitignored): `SmokeTest.ipynb` runs eleven automated checks — including a full MECLib model run — and prints `SMOKE_OK`; `MiniTest.ipynb` is the minimal widget-pattern test. Build with `-contents content -contents tests/smoke` to include them locally; the deploy workflow builds `-contents content` only, so they can never reach the public site.
- This document: `docs/instructor_guide.tex`, compiled with `tectonic instructor_guide.tex`. A plain-text quick reference lives at the repository root (`INSTRUCTOR_GUIDE.md`); when one changes, update the other.